

Task 16 — Hypothesis Testing & A/B Tests

Data Analyst · Phase 1 Industry Immersion · Task Brief

DATA ANALYST

PlaceMux · Phase 1 · Task Brief

DIFFICULTY

Advanced · Advanced

Level 16 of 20 · ramp 76%

Objective

What this task is: Design and run hypothesis tests / A-B experiments to decide whether an observed difference is real or chance.

Why it matters: Decisions based on noise waste money. Proper testing separates real effects from random fluctuation.

What you will deliver

- A documented test/experiment with effect size, significance and a recommendation.

Inputs & prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.
- Required tools for this task: scipy.stats / statsmodels, pandas, a power calculator

Step-by-step execution

Work through these in order. Don't skip the verification steps — that's where 'looks done' becomes 'is done'.

1. State the hypothesis and the decision it informs.
2. Choose the right test and significance level up front.
3. Estimate required sample size/power.
4. Run the test on clean, comparable groups.
5. Report effect size with the p-value and confidence interval.
6. Translate the result into a clear recommendation.

Tools & stack

- scipy.stats / statsmodels
- pandas
- a power calculator

Definition of Done

You are done when all of these are true and demoable

- A documented test/experiment with effect size, significance and a recommendation.
- Demonstrable live on real (even if small) data, not just described.

Pitfalls to avoid

Worry if any of these are true of your work

- p-hacking by testing until something sticks.
- Significant but trivially small effects.
- Unequal or contaminated groups.

Where this fits in the track

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.